Low-profile SIW slot array with characteristic-mode-guided decoupling for 28 GHz 5G mmWave front ends: design in an open-source hybrid FDTD/MoM environment

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Abstract

A dual-port, low-profile ($0.054\lambda_0$ thick) substrate-integrated-waveguide (SIW) longitudinal-slot array for the 28 GHz 5G mmWave band is designed, and the mutual coupling between its two 1×5 sub-arrays is reduced with a row of parasitic cavity-backed slots whose length and position are dimensioned with characteristic-mode analysis (CMA). The whole design is carried out in an open-source environment written for this work: a Numba-accelerated 3-D finite-difference time-domain (FDTD) solver with convolutional perfectly matched layers, TE_{10} SIW ports and an aperture near-to-far-field transformation, and a Rao–Wilton–Glisson method-of-moments (MoM) solver with a Harrington–Mautz characteristic-mode eigensolver in both the electric-current (finite conductor) and the magnetic-current (aperture) formulations. The FDTD solver reproduces the analytical SIW dispersion within 1.9 % at 28 GHz. Full-wave single-slot characterisation on the 0.508 mm RT/duroid 5880 substrate shows a resonant conductance $g = K\sin^2(\pi x_0/a)$ with $K = 17.2$, more than an order of magnitude above the thick-waveguide Stegen value, so that a five-slot standing-wave sub-array is matched with offsets of only 0.2 mm. The aperture (magnetic-current) formulation of CMA shows why slits etched in the shared top wall between closely spaced SIWs cannot decouple them: over the sub-cut-off channel between the via rows an aperture is backed by a virtual short and never resonates, which

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the FDTD confirms for slit lengths from 3.6 mm to a continuous 24 mm slit (isolation unchanged at -15.3 dB for a 7.0 mm spacing). Widening the channel into via-bounded parasitic cavities and offsetting the slots so that they couple to the cavity mode yields a resonant parasitic row: for a 9.4 mm ($0.88\lambda_0$) sub-array spacing the 28 GHz isolation improves from -17.5 dB to -23.5 dB (5.9 dB), the worst isolation over 27.5–28.5 GHz to -20.5 dB, the realised gain rises from 10.1 dBi to 11.9 dBi and the envelope correlation coefficient stays below 0.033, with no additional layer or component. All solvers, scripts and data are released, so the design is reproducible without commercial electromagnetic software.

Keywords: substrate integrated waveguide, slot array, characteristic modes, mutual coupling, decoupling, 28 GHz, 5G mmWave, FDTD, method of moments, open-source

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1. Introduction

Millimetre-wave (mmWave) bands around 28 GHz are central to 5G New Radio because of the contiguous spectrum they offer, but the free-space path loss and the blockage sensitivity of these bands require directive, electronically steerable antennas at both the base station and the user terminal [1, 2]. The front-end antenna must therefore be a compact array that is thin enough to be embedded in a handset or in a tile of a massive-MIMO panel, that integrates with the printed-circuit-board (PCB) feed of the transceiver, and that offers sufficient port-to-port isolation for the sub-arrays to be driven independently [3, 4].

Substrate integrated waveguide (SIW) technology provides exactly this combination of low profile, low loss and PCB compatibility at mmWave frequencies [5, 6, 7, 8]. Longitudinal-slot arrays cut in the broad wall of an SIW are the planar counterpart of the classical waveguide slot arrays and can be synthesised with Elliott's procedure once the admittance of an isolated slot is known [9, 10, 11, 12, 13]. Several SIW slot arrays have been reported for the 28 GHz band [14, 15, 16, 17, 18], usually as single-port arrays fed by a corporate SIW divider. When two SIW sub-arrays are placed side by side on the same substrate, their slots are parallel and displaced in their E-plane, the strongest-coupling arrangement for slots in a common ground plane, and the port isolation is only 15–20 dB for a centre spacing of 0.6–$0.9\lambda_0$. Reported remedies include metamaterial or electromagnetic-bandgap shields [19, 20],

indirect-coupling feed networks [21], via chokes and field-confining structures [4]; they are either bulky (additional layers or periodic cells) or specific to one feed topology.

The theory of characteristic modes (TCM) [22, 23, 24, 25, 26] has become a systematic tool for decoupling, because the eigencurrents of the shared structure reveal which mode carries energy from one port to the other and how the structure must be perturbed to suppress it [27, 28, 29, 30, 31, 32]. Most of these works address sub-6 GHz terminal antennas on finite ground planes, and they use the electric-current formulation. For slot radiators in a common ground plane the natural formulation is the dual, magnetic-current (aperture) one, and its application to the shared top wall of SIW slot arrays at mmWave frequencies has, to the best of our knowledge, not been reported. A second practical obstacle is that such designs are normally tied to commercial full-wave software, so that the published results cannot be reproduced by groups without those licences.

This paper addresses both issues. Its contributions are:

1. an open-source electromagnetic design environment, written from scratch in Python with Numba acceleration [33], comprising a 3-D FDTD solver [34, 35] with SIW waveguide ports and an aperture near-to-far-field transformation, an RWG method-of-moments solver [36] with a characteristic-mode eigensolver in both the electric-current and the magnetic-current formulation, and the analytical SIW and slot-array synthesis equations; the solvers are validated against the analytical SIW dispersion relation and the resonances of a strip dipole and of a slot;
2. the full-wave characterisation of a longitudinal slot in a thin ($h = 0.047\lambda_0$) SIW at 28 GHz, showing that the resonant conductance per slot is more than an order of magnitude larger than predicted by the Stegen formula and that the slot resonance shifts strongly with the offset, and the resulting Elliott synthesis of a 1×5 standing-wave sub-array with 0.2 mm offsets;
3. a characteristic-mode analysis of the coupling in the aperture formulation, which explains, and the FDTD confirms, that slits etched in the top wall over the sub-cut-off channel between adjacent SIWs cannot decouple them, whatever their length;
4. a characteristic-mode-guided decoupling structure, a row of offset parasitic slots backed by via-bounded cavities between the sub-arrays, that

raises the 28 GHz isolation by 5.9 dB and the realised gain by 1.8 dB at no cost in profile.

Section 2 describes the design environment and its validation, Section 3 the sub-array synthesis, Section 4 the characteristic-mode analysis of the coupling and the design of the parasitic row, Section 5 the full-wave results and a comparison with the literature, and Section 6 the conclusions.

2. Open-source design environment

2.1. Architecture

The environment (`siwlab`, released with this paper) is organised in five modules. `siw` implements the equivalent-width, cutoff, dispersion and attenuation relations of the SIW; `slots` implements the Elliott standing-wave synthesis with full-wave slot data; `mesh` and `mom` implement the triangular mesher, the RWG electric-field integral equation (EFIE), its magnetic-current dual for apertures in a ground plane, and the characteristic-mode eigensolver; `fdtd` and `geometry` implement the 3-D FDTD solver and the parametric SIW array builder. All computations reported here were run on four CPU cores; the longest single FDTD run takes about ten minutes.

2.2. SIW model

For via diameter d, pitch p and via-row spacing a the equivalent rectangular-waveguide width is $a_{\mathrm{eff}} = a - d^2/(0.95p)$ [7], and the TE_{10} phase constant is $\beta = \sqrt{k_0^2\varepsilon_r - (\pi/a_{\mathrm{eff}})^2}$. The design rules $p \le 2d$ and $d < \lambda_c/5$ [5, 6] are checked automatically. The substrate is Rogers RT/duroid 5880 ($\varepsilon_r = 2.2$, $\tan\delta = 0.0009$, $h = 0.508\,\mathrm{mm}$) with $d = 0.4\,\mathrm{mm}$, $p = 0.8\,\mathrm{mm}$ and $a = 5.6\,\mathrm{mm}$, giving $a_{\mathrm{eff}} = 5.39\,\mathrm{mm}$ and $f_c = 18.75\,\mathrm{GHz}$, i.e. $f_c/f_0 = 0.67$ at 28 GHz.

2.3. FDTD solver

The FDTD solver uses a nonuniform Yee grid (0.1 mm cells in the SIW and slot region, graded to 0.2 mm in the margins and to 0.25 mm in the air above the array; five cells across the substrate), convolutional PML (CPML) [35] on the lateral and upper boundaries, a perfect electric conductor (PEC) on the lower boundary (ground plane), cell-averaged permittivity and conductivity on the field edges, and edge-based PEC masks for the top metal, the vias (cell-centred staircase that conserves the via cross-section) and the

shorting walls. The via rows are extended into the PML so that the SIW feed lines are matched. A TE_{10} mode is launched as a soft E_z source with the profile $s(x) = \sin[\pi(x - x_1)/a_{\mathrm{eff}}]$ and the modal voltage and current, $V(t) = \iint E_z\, s\,\mathrm{d}S$ and $I(t) = \iint H_x\, s\,\mathrm{d}S$, are recorded on reference planes. The S-parameters follow from the ratio of the spectra to those of a slot-free reference line with identical source and grid, which also yields the incident power $P_{\mathrm{inc}} = \mathrm{Re}(VI^*)/(2N)$ with $N = \iint s^2\,\mathrm{d}S$. The tangential field in every aperture of the top wall is accumulated by a running discrete Fourier transform, and the far field is obtained from the equivalent magnetic current $\mathbf{M} = -2\hat{\mathbf{z}} \times \mathbf{E}_a$ on the infinite PEC plane [37]; the radiated power from the hemisphere integral of the radiation intensity gives the radiation efficiency, and the envelope correlation coefficient (ECC) is computed from the embedded far fields of the two ports [38]. The excitation is a Gaussian-modulated pulse centred at 28 GHz with a 1/e half-bandwidth of 6 GHz; the time step is $\Delta t = 0.1885\,\mathrm{ps}$ (Courant number 0.99) and the simulated interval is 2.6 ns for the single-slot and 3.6 ns for the array models. Numba just-in-time compilation with parallel loops gives 10 ms per time step for 2.3×10^6 cells.

2.4. MoM and characteristic-mode solver

The EFIE is discretised with RWG basis functions on a structured triangular mesh whose lines coincide with every aperture edge. The impedance matrix

$$Z_{mn} = \mathrm{j}\omega\mu \iint \mathbf{f}_m \cdot \mathbf{f}_n G\,\mathrm{d}S\,\mathrm{d}S' - \frac{1}{\mathrm{j}\omega\varepsilon} \iint (\nabla\cdot\mathbf{f}_m)(\nabla\cdot\mathbf{f}_n) G\,\mathrm{d}S\,\mathrm{d}S' \tag{1}$$

is assembled triangle-pair-wise with a six-point Gauss rule on the test triangle and a 4^4 (4^5 for the self term) barycentric subdivision of the source triangle for near interactions, in parallel Numba kernels. The characteristic modes are the solutions of $\mathbf{X}\mathbf{J}_n = \lambda_n \mathbf{R}\mathbf{J}_n$ [23], computed after projecting onto the subspace of $\mathbf{R}$ with eigenvalues above 10^{-9} of its maximum, and tracked over frequency by the $\mathbf{R}$-weighted correlation of the eigencurrents. The modal significance is $\mathrm{MS}_n = |1 + \mathrm{j}\lambda_n|^{-1}$ and the characteristic angle $\alpha_n = 180^\circ - \arctan\lambda_n$ [24]. For apertures in an infinite ground plane the dual (magnetic-current) formulation is used: with the image theorem the aperture admittance operator of each half-space i is

$$Y^{(i)}_{mn} = 2\Big[\mathrm{j}\omega\varepsilon_i \iint \mathbf{f}_m \cdot \mathbf{f}_n G_i\,\mathrm{d}S\,\mathrm{d}S' - \frac{1}{\mathrm{j}\omega\mu_0} \iint (\nabla\cdot\mathbf{f}_m)(\nabla\cdot\mathbf{f}_n) G_i\,\mathrm{d}S\,\mathrm{d}S'\Big], \tag{2}$$

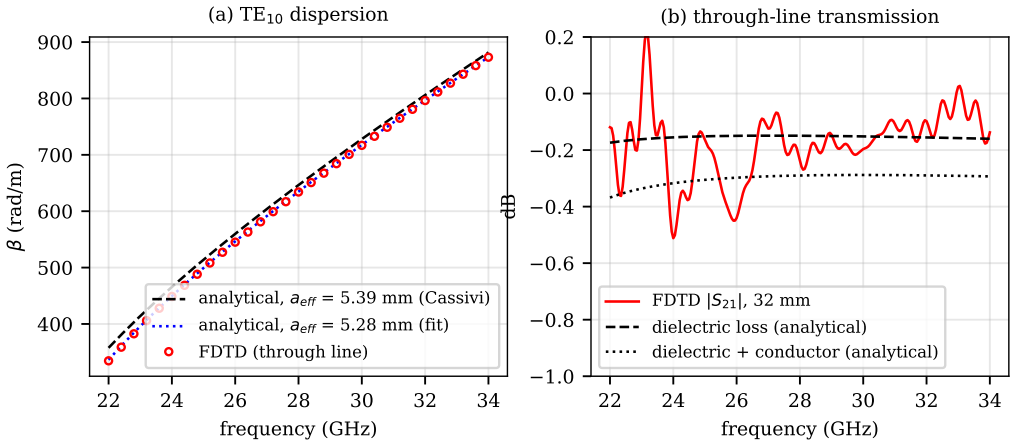


Figure 1: Validation of the FDTD solver on an SIW through line: (a) TE_{10} phase constant versus the analytical model; (b) transmission over 32 mm versus the analytical dielectric and conductor losses.

and the aperture modes are the eigenvectors of $\mathbf{BM}_n = \lambda_n \mathbf{GM}_n$ with $\mathbf{Y} = \mathbf{Y}^{(1)} + \mathbf{Y}^{(2)} = \mathbf{G} + \mathrm{j}\mathbf{B}$. The solver was verified on a $10\,\mathrm{mm}\times0.5\,\mathrm{mm}$ strip, whose first characteristic mode resonates ($\lambda_1 = 0$) at $L = 0.457\lambda$, and on a $4.0\,\mathrm{mm}\times0.4\,\mathrm{mm}$ aperture in air, which resonates at $L = 0.46\lambda$, both in agreement with the known dipole and slot resonances.

2.5. Validation on an SIW through line

Fig. 1 compares the TE_{10} phase constant extracted from the FDTD transmission phase of a 32 mm long SIW with the analytical relation. The agreement is within 1.9 % at 28 GHz; the FDTD data are reproduced exactly by the analytical model with $a_{\mathrm{eff}} = 5.28$ mm, i.e. 2.1 % below the Cassivi value, which is the expected effect of the staircase representation of the vias (radius of two cells). The guided wavelength used in the synthesis is therefore the full-wave value $\lambda_g = 9.91$ mm at 28 GHz. The through-line transmission of -0.24 dB at 28 GHz is consistent with the analytical dielectric loss of 0.15 dB and a residual leakage through the via rows at the limit $p = 2d$ of the design rule.

3. Sub-array synthesis

3.1. Geometry

The antenna (Fig. 2) consists of two parallel SIWs with centre spacing D, each carrying five longitudinal slots of width $w = 0.4\,\mathrm{mm}$, alternately offset by $\pm x_0$ from the SIW axis, spaced by $d_s = \lambda_g/2$ and terminated by a shorting via wall at $\lambda_g/4$ from the last slot, i.e. a standing-wave array [10]. The two SIWs are fed at the opposite end (ports P1 and P2). The region between the inner via rows, of width $D - a$, is covered by the common top metal; in the final design it is partitioned by via posts into cavities that back the parasitic decoupling slots (Section 4). The total thickness is the substrate plus two copper layers, $\approx 0.58\,\mathrm{mm}$ ($0.054\lambda_0$). Two spacings are studied: $D = 7.0\,\mathrm{mm}$ ($0.65\lambda_0$, a 1.4 mm channel) and $D = 9.4$ mm ($0.88\lambda_0$, a 3.8 mm channel).

3.2. Single-slot characterisation

Elliott's procedure requires the resonant length and the resonant conductance of an isolated slot as functions of its offset. Both were obtained with the FDTD solver from a single slot in a matched SIW through line: the shunt admittance normalised to the TE_{10} characteristic admittance is $y = g + \mathrm{j}b = 2(1 - S_{21})/S_{21}$, where S_{21} is referred to the slot-free line. Fig. 3(a) shows $y(f)$ for $x_0 = 0.2\,\mathrm{mm}$ and several lengths, Fig. 3(b) the resonant frequency ($b = 0$) versus length for three offsets and Fig. 3(c) the resonant conductance versus offset. Slot lengths are quoted in the on-grid convention of the FDTD model (aperture edges on grid lines, edge nodes metallic), which gives a length resolution of 0.2 mm. Two features distinguish the thin SIW from the classical thick waveguide. First, the resonant conductance is very large: the least-squares fit $g_r = K\sin^2(\pi x_0/a)$ gives $K = 17.2$, whereas the Stegen formula $g_r = 2.09(a/b)(\lambda_g/\lambda_0)\cos^2(\pi\lambda_0/2\lambda_g)\sin^2(\pi x_0/a)$ [10] with $b = h$ gives $K \approx 0.5$, because its $\cos^2$ factor is derived for $\lambda_g > \lambda_0$ and is not applicable to the dielectric-filled guide where $\lambda_g < \lambda_0$. Second, the resonant frequency depends strongly on the offset (27.9 GHz at $x_0 = 0.5\,\mathrm{mm}$ but 26.4 GHz at $x_0 = 0.2\,\mathrm{mm}$ for $L_s = 4.6\,\mathrm{mm}$), because the internal reactive loading by the guide is comparable to the external slot reactance; the slot data must therefore be taken at the offset actually used. At $x_0 = 0.2\,\mathrm{mm}$ the interpolated 28 GHz resonant length of the isolated slot is 4.12 mm and $g_r \approx 0.20$.

3.3. Elliott synthesis

For a uniform standing-wave array the slot conductances are $g_n = 1/N$ so that the input is matched ($\sum g_n = 1$) [9, 10]. With $K = 17.2$, $g_n = 0.2$ requires $x_0 = 0.19\,\text{mm}$; the offset is rounded to the 0.2 mm grid of the FDTD model ($\sum g_n = 1.08$, i.e. $|S_{11}| \approx 0.04$ at resonance). A five-slot sub-array is therefore the natural choice for this substrate: a four-slot array would need $g_n = 0.25$ and an offset of 0.22 mm, while the larger offsets of a conventional design would over-couple the slots. The slot spacing $d_s = \lambda_g/2 = 4.96\,\text{mm}$ and the distance from the short, $\lambda_g/4 = 2.48\,\text{mm}$, were rounded to 5.0 mm and 2.5 mm. The first full-wave run of the complete array with the on-grid length $L_s = 4.0\,\text{mm}$ showed the band centred at 27.5 GHz, because the external mutual coupling between the five slots lowers their resonance by about 0.9 GHz with respect to the isolated slot; a single iteration with the next shorter on-grid length, $L_s = 3.8$ mm, centres the band at 28.5 GHz (Section 5). This two-step flow, first-order Elliott synthesis followed by one full-wave correction, replaces the iterative second-order procedure of [9] at a cost of one array simulation.

4. Characteristic-mode-guided decoupling

4.1. Aperture-system view of the coupling

By the equivalence principle, the field above a conducting plane with apertures is produced entirely by the magnetic currents in the apertures and their images; the surface current on the plane is a consequence of that field, not its source. For two slot sub-arrays in a common top wall the mutual coupling is therefore an interaction between apertures, and the appropriate characteristic-mode description is the magnetic-current one, (2). This has two consequences that guide the design. (i) Any structure etched in the top wall decouples the sub-arrays only in so far as it is itself an aperture carrying a magnetic current, i.e. a parasitic radiator; merely interrupting the surface current path is ineffective. (ii) A parasitic aperture is effective only near its own resonance, which depends on what backs it: the aperture admittance is the sum of the exterior (air) admittance and the interior admittance of the region below the top wall.

The exterior part is modelled with (2) using an effective permittivity for the lower half-space, calibrated once from the full-wave slot data: the isolated radiating slot that resonates at 28 GHz in the FDTD model ($L_s = 4.12$ mm, $x_0 = 0.2\,\text{mm}$) has its aperture-mode resonance ($\alpha_1 = 180^\circ$) at 33.3 GHz

in air and at 28 GHz for $\varepsilon_{\text{eff}} = 1.88$ (Fig. 4(a)). With this medium the aperture-mode resonance of a 0.3 mm wide parasitic slot is computed as a function of its length (Fig. 4(b)); a 28 GHz characteristic resonance requires $L_d = 3.96$ mm. The coupling between the sub-arrays is then quantified with the full aperture system: ten radiating slots plus the parasitic apertures, with the parasitic unknowns eliminated by a Schur complement, $\mathbf{Y}_{\text{eff}} = \mathbf{Y}_{ss} - \mathbf{Y}_{sp}\mathbf{Y}_{pp}^{-1}\mathbf{Y}_{ps}$, and the normalised mutual admittance $|y_{12}|/|y_{11}|$ evaluated between the dominant aperture modes of the two sub-arrays. Fig. 5(a) shows that a row of five parasitic slots of the right length reduces this external mutual admittance by 10–20 dB, with the strongest cancellation close to the aperture-mode resonance of the parasitic slot, for both spacings; Fig. 5(b) shows the modal significance of the aperture-system modes, among which the modes with a large current fraction in the parasitic slots become significant at 28 GHz.

4.2. Why top-wall slits over the channel do not decouple

The exterior model assumes that the aperture is backed by an open region. In the compact configuration ($D = 7.0$ mm) the channel between the inner via rows is only 1.4 mm wide (1.0 mm clear) and 0.508 mm high; its lowest mode cuts off above 100 GHz, so a slit etched over it sees, on its lower side, an evanescent region that is equivalent to a short circuit at a depth of $0.05\lambda_0$. The interior admittance is then a very large inductive susceptance and the slit cannot resonate anywhere near 28 GHz. The FDTD confirms this (Table 1): five slits of 3.6, 4.0, 4.4 or 4.8 mm or a single continuous 24 mm slit at the channel centre leave the isolation at -15.3 dB within $\pm$1.5 dB, while the aperture model with an open backing would have predicted a reduction. The electric-current CMA of the finite top wall gives the same warning from the other side: the slits reduce the gap-crossing index of the dominant modes from 1.17 to 0.36 (Fig. 6), i.e. they do interrupt the transverse surface current between the sub-arrays, but since that current is not the source of the exterior field the isolation does not change. Interrupting the surface current is therefore not a valid design criterion for slot radiators in a common ground plane.

4.3. Parasitic cavity-backed slots

The parasitic aperture must be backed by a region that supports a resonant field. The channel is therefore widened to 3.8 mm ($D = 9.4$ mm) and partitioned by via posts at the slot pitch into five cavities of $3.8 \times 5.0\,\text{mm}^2$

whose TE_{101}-type mode, loaded by the slot, lies close to the band. A slot on the cavity axis does not couple to this mode (its transverse surface current vanishes on the axis, exactly as for a centred slot in an SIW), so the parasitic slots are offset by $x_d = 0.8$ mm from the channel axis; their length is the CMA value $L_d = 3.96$ mm, implemented as 4.0 mm on the 0.2 mm length grid. The FDTD verification (Table 1) shows the expected behaviour: with the cavities but a centred slot the isolation is unchanged, whereas the offset parasitic row improves the 28 GHz isolation from -17.5 dB to -23.5 dB; shorter or longer slots and a larger offset are less effective, so the CMA length is within the 0.2 mm resolution of the full-wave optimum. The parasitic slots also radiate in phase with the array, which is why the gain increases (Section 5).

5. Full-wave results and discussion

5.1. S-parameters

Fig. 7 shows the FDTD reflection and transmission coefficients of the compact array ($D = 7.0$ mm, no parasitic structure) and of the final array ($D = 9.4$ mm) without and with the parasitic row. The compact array has an isolation of -15.3 dB at 28 GHz (worst value over 27.5–28.5 GHz, the n261 band, -14.9 dB). Widening the spacing alone improves the isolation to -17.5 dB (worst -17.2 dB). The CM-guided parasitic row improves it further to -23.5 dB at 28 GHz and to -20.5—26.2 dB over 27.5–28.5 GHz, i.e. by 5.9 dB at the centre frequency. The input match of the standing-wave sub-array is marginal in all configurations (minimum $|S_{11}|$ of -10.4 dB without and -11.3 dB with the parasitic row; -10 dB band 28.23–28.68 GHz and -6 dB band 27.77–29.07 GHz, 4.6 %), because the first-order synthesis was corrected only once for the slot-to-slot coupling and the slot length is restricted to the 0.2 mm grid; a second-order Elliott iteration on a finer grid would deepen the match without changing the decoupling mechanism. Table 1 summarises all configurations simulated.

5.2. Radiation

Fig. 8 shows the embedded gain patterns of port 1 at 28 GHz in the two principal planes, the realised gain versus frequency and the ECC computed from the embedded patterns of the final array. The sub-array radiates a broadside beam with a realised gain of 11.9 dBi at 28 GHz with the parasitic row (10.1 dBi without; the parasitic slots are excited in phase and enlarge the radiating aperture), a half-power beamwidth of about 23° in the y-z

Table 1: FDTD results for all configurations (port 1 excited; $|S_{11}|$, $|S_{21}|$ and the realised gain G_r at 28 GHz; $|S_{21}|_{\max}$ is the worst isolation over 27.5–28.5 GHz; x_d in mm).

D (mm)	structure	L_d (mm)	$\lvert S_{11}\rvert$	$\lvert S_{21}\rvert$	$\lvert S_{21}\rvert_{\max}$	G_r (dBi)
7.0	none	–	-9.0	-15.3	-14.9	10.9
7.0	slit	3.6	-9.3	-14.8	-14.4	10.8
7.0	slit	4.0	-9.4	-14.7	-14.3	10.8
7.0	slit	4.4	-9.4	-14.5	-14.2	10.8
7.0	slit	4.8	-9.5	-14.4	-14.0	10.7
7.0	cont. slit	24.0	-10.0	-13.9	-13.5	10.6
9.4	none	–	-9.1	-17.5	-17.2	10.1
9.4	cav. slot (centred)	3.6	-9.0	-17.0	-16.8	10.8
9.4	cav. slot (x_d=0.8)	3.6	-7.3	-17.8	-17.6	10.9
9.4	cav. slot (centred)	4.0	-8.5	-17.4	-17.3	11.2
9.4	cav. slot (x_d=0.8)	4.0	-8.2	-23.5	-20.5	11.9
9.4	cav. slot (x_d=0.8)	4.4	-8.9	-20.3	-19.1	11.5

plane (five slots) and 62° in the x-z plane, and a radiation efficiency of 96 % (dielectric loss only; the copper is modelled as PEC, so that the conductor loss of about 0.11 dB estimated analytically for the 25 mm slot section should be subtracted). Without the parasitic row the coupling tilts the beam of the excited sub-array by +13° in the x-z plane; with it the tilt is +7°. The ECC between the two ports, computed from the embedded patterns, is 0.014 without and 0.033 with the parasitic row at 28 GHz: it remains far below the usual 0.5 limit, the slight increase reflecting the fact that the parasitic slots are shared by both ports and radiate with either of them.

5.3. Comparison with the literature

Table 2 compares the proposed antenna with reported 28 GHz SIW slot arrays and decoupled mmWave arrays. The proposed design combines the lowest profile with an isolation comparable to that of dedicated decoupling networks [21] or shields [19, 20], using no additional layer or component, and the parasitic row adds gain instead of absorbing it. Its bandwidth is that of a standing-wave array; a travelling-wave or centre-fed configuration would widen it at the cost of a frequency-scanning beam, which is left for future work.

5.4. Limitations

The results are numerical. The FDTD model uses PEC metal, an infinite ground plane and a 0.1 mm staircase grid; the via representation shifts a_{eff}

Table 2: Comparison with reported mmWave SIW slot arrays and decoupled arrays (simulated results for this work).

Ref.	f_0 (GHz)	Profile (λ_0)	Gain (dBi)	Isolation (dB)	Decoupling
[14]	28/38	0.047	11.9	–	single port
[15]	28/38	0.05	9.3–10.9	–	single port
[17]	28	0.05	12.3	–	single port
[18]	28	0.07	4.7/8	> 30	port placement (annular slot)
[21]	10	0.05	–	35	two-layer SIW coupler network
[19]	60	0.15	10.1	≈ 25	metamaterial shield
[20]	28	> 0.1	9–13.3	> 40	stacked EBG + DGS
This work	28	0.054	11.9	-20.5 to -26.2 (27.5–28.5 GHz)	CM-guided parasitic cavity-backed slots, same layer

by 2.1 %, which was compensated by using the full-wave λ_g in the synthesis. The aperture-mode CMA uses a single effective permittivity calibrated from one full-wave slot resonance and an open backing; it gives the resonant length of the parasitic slot but not the cavity loading, so the final offset and length were confirmed by a short full-wave sweep, as done here. The input match of the standing-wave sub-array (-11 dB minimum) would be improved by a second-order Elliott iteration including the mutual coupling. A microstrip-to-SIW transition and a measurement on a fabricated prototype are the natural next steps; the released environment allows both to be simulated without proprietary software.

6. Conclusion

A dual-port, low-profile SIW slot array for 28 GHz 5G front ends has been designed entirely with an open-source hybrid FDTD/MoM environment developed for this work. The characteristic-mode analysis in the aperture (magnetic-current) formulation showed that decoupling slot radiators in a common top wall requires a resonant parasitic aperture with an open backing, which rules out slits over the sub-cut-off channel of closely spaced SIWs, and led to a row of offset, cavity-backed parasitic slots that raises the port isolation by 5.9 dB at 28 GHz while adding 1.8 dB of gain and no thickness. The design flow, from the SIW equations through the full-wave slot charac-

terisation, the Elliott synthesis, the CMA-based parasitic dimensioning and the full-wave verification, is reproducible from the released scripts, and the same environment applies directly to larger tiles, dual-polarised SIW arrays and other mmWave bands.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

The complete source code of the design environment, all scripts and all simulation data are available in the repository accompanying this article.

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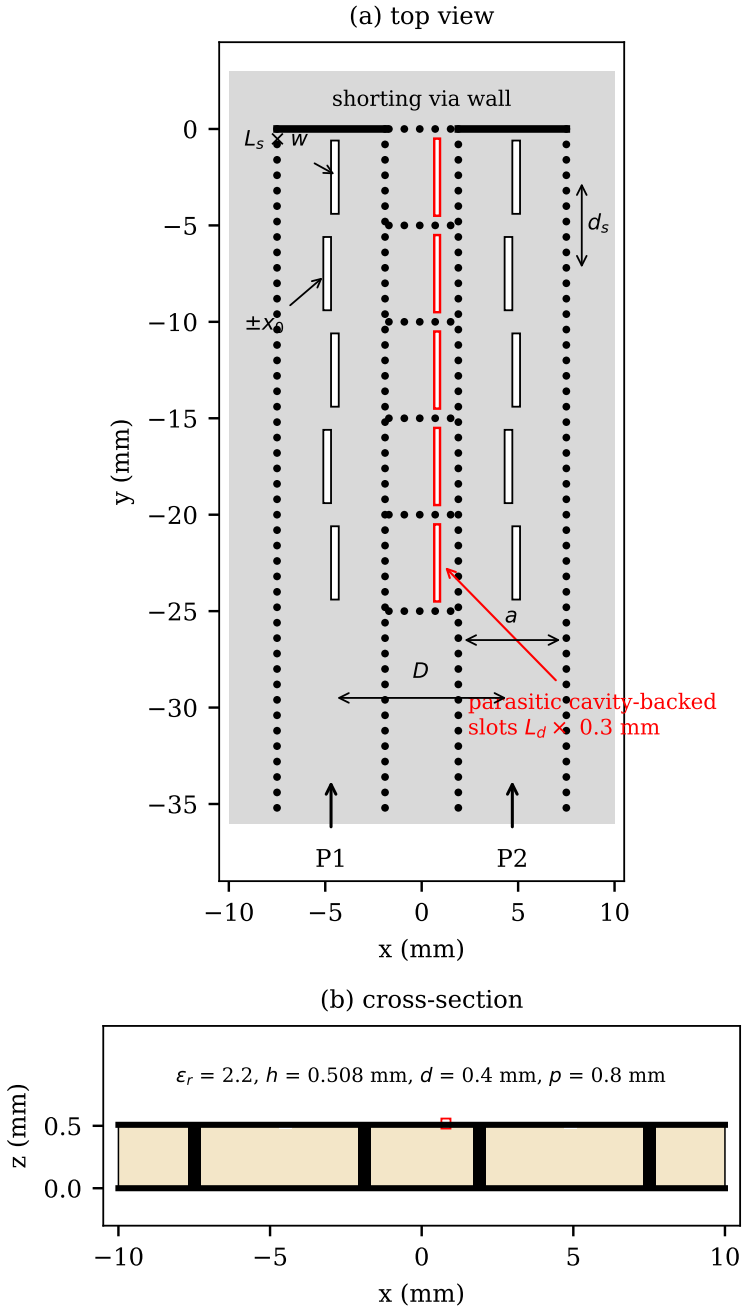


Figure 2: Dual-port SIW slot array with the CM-guided parasitic row (final design, $D = 9.4$ mm): (a) top view; (b) cross-section.

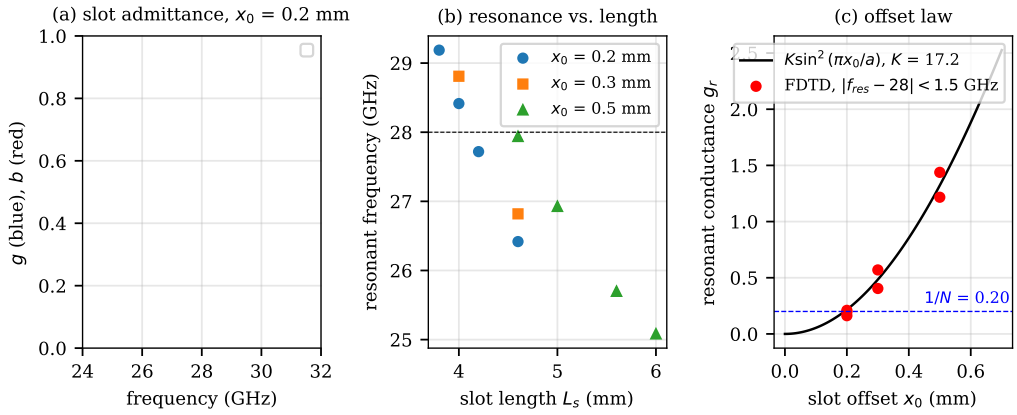


Figure 3: Full-wave characterisation of a longitudinal slot in the 0.508 mm SIW: (a) normalised admittance for $x_0 = 0.2$ mm; (b) resonant frequency versus slot length; (c) resonant conductance versus offset with the $\sin^2$ fit; the dashed line is the conductance $1/N$ required for a matched N-slot standing-wave array.

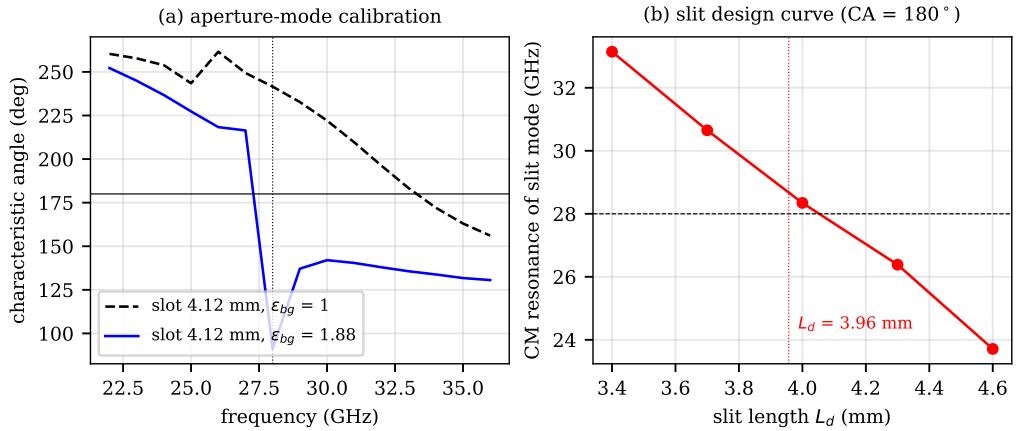


Figure 4: Aperture-mode (magnetic-current) characteristic-mode analysis: (a) characteristic angle of the dominant mode of the radiating slot in air and in the calibrated effective medium; (b) resonance of a 0.3 mm wide parasitic slot versus its length.

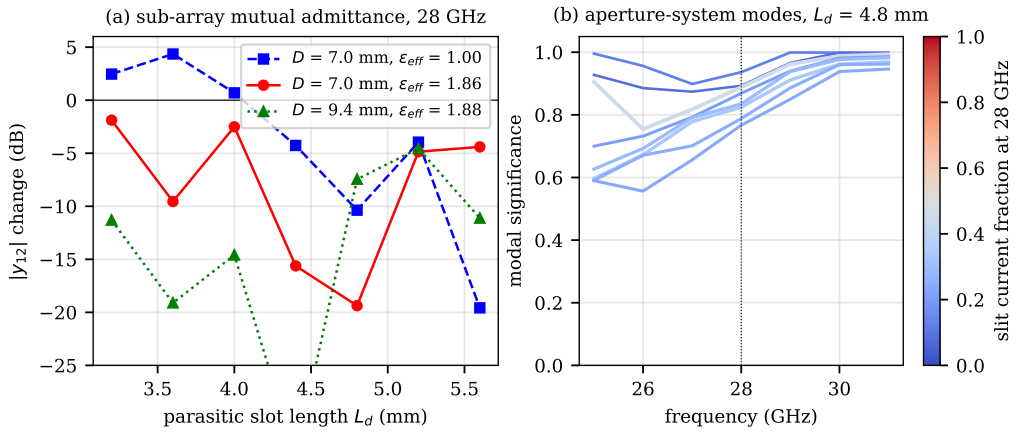


Figure 5: Aperture-system analysis at 28 GHz: (a) change of the external mutual admittance between the sub-arrays produced by a row of five parasitic apertures of length L_d (Schur complement of the aperture admittance matrix); (b) modal significance of the aperture-system modes for $D = 7.0\,\text{mm}$ and $L_d = 4.8\,\text{mm}$, coloured by the fraction of the modal current located in the parasitic apertures.

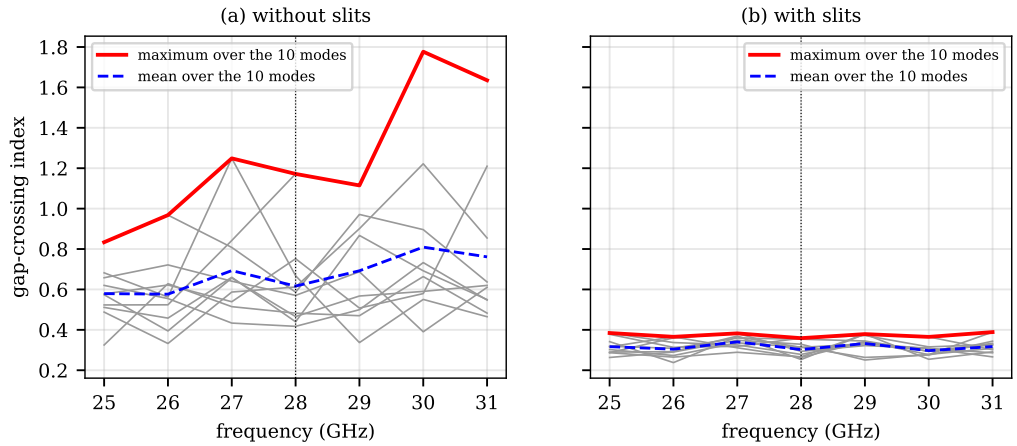


Figure 6: Electric-current CMA of the finite shared top wall ($D = 7.0\,\text{mm}$): gap-crossing index of the ten lowest-$|\lambda|$ modes (a) without and (b) with five 4.0 mm slits; the slits suppress the transverse surface current between the sub-arrays without affecting the isolation.

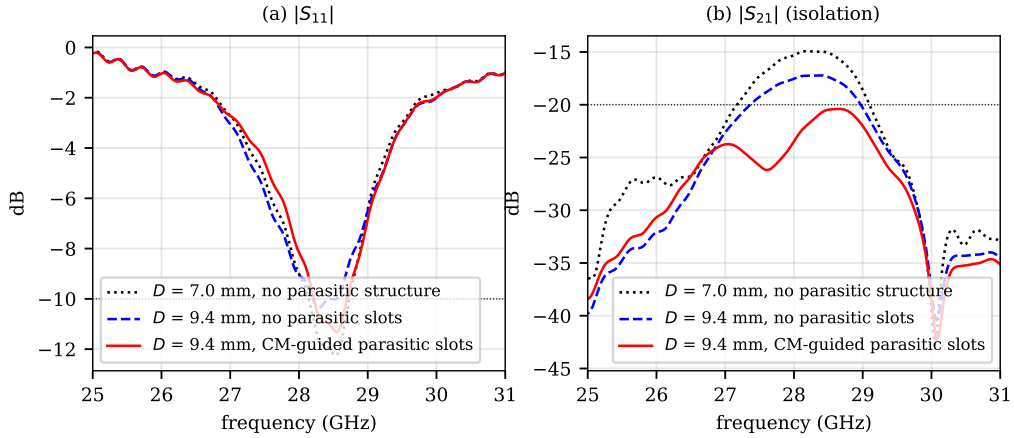


Figure 7: FDTD S-parameters of the dual-port array: compact spacing without decoupling, and final spacing without and with the CM-guided parasitic row: (a) reflection; (b) isolation.

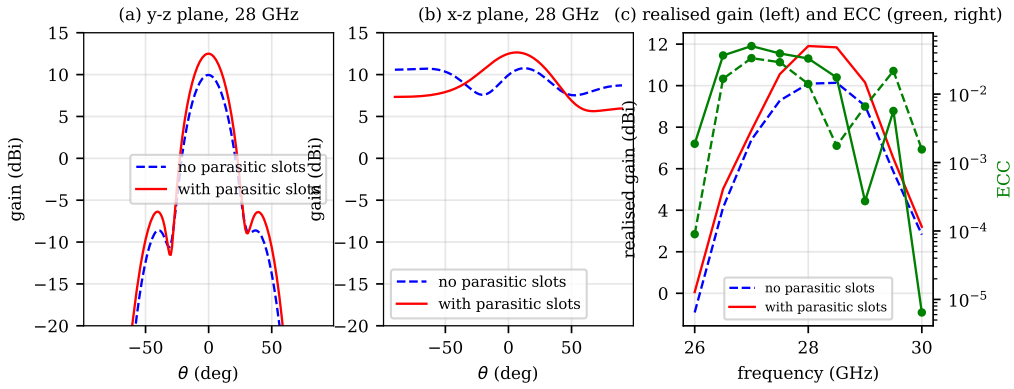


Figure 8: FDTD radiation results for port 1 of the final array: (a) y-z plane and (b) x-z plane gain patterns at 28 GHz; (c) realised gain and ECC versus frequency.